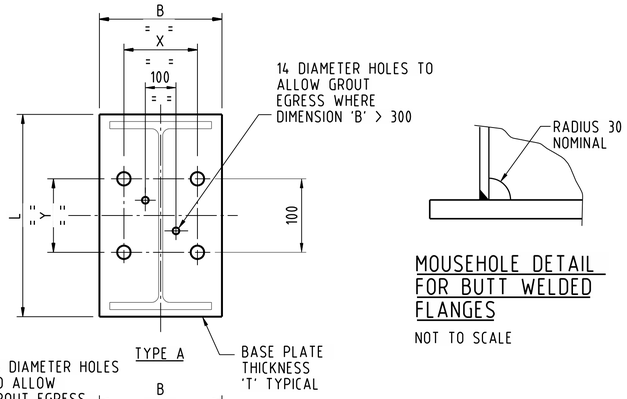
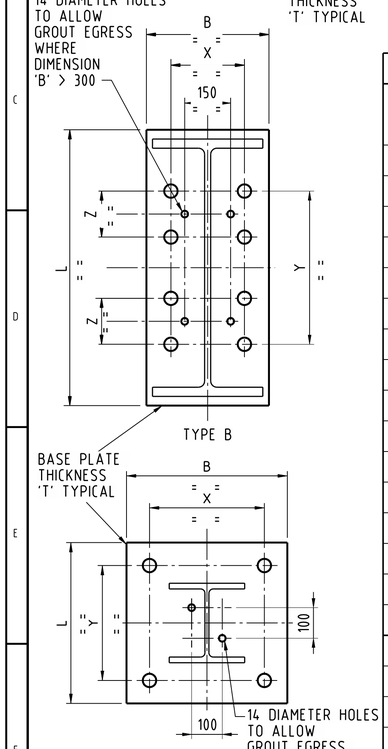
MA'ADEN PHOSPHATE PROJECT-A1 BORDER-REV A02



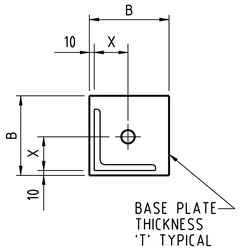

## MOUSEHOLE DETAIL FOR BUTT WELDED FLANGES

NOT TO SCALE

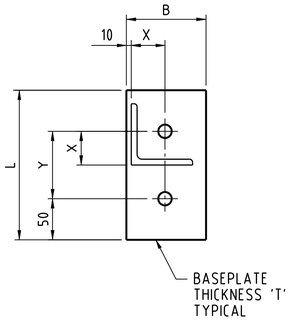



| MEMBER | DIMENSIONS | | | | | | |
|---|---|---|---|---|---|---|---|
| | TYPE | B | L | X | Y | Z | T |
| W8x18 | D | 200 | 250 | 120 | - | - | 16 |
| W10x22 | A | 200 | 300 | 120 | 120 | - | 20 |
| W10x26 | A | 200 | 300 | 120 | 120 | - | 20 |
| W12x26 | A | 200 | 350 | 120 | 180 | - | 20 |
| W14x34 | A | 200 | 400 | 120 | 200 | - | 20 |
| W16x40 | A | 230 | 450 | 150 | 250 | - | 20 |
| W18X46 | A | 200 | 500 | 120 | 300 | - | 25 |
| W21x50 | A | 200 | 580 | 120 | 350 | - | 25 |
| W24x68 | A | 270 | 650 | 150 | 450 | - | 32 |
| W24x84 | A | 270 | 650 | 150 | 450 | - | 32 |
| W27x94 | B | 300 | 750 | 190 | 500 | 150 | 32 |
| W30x116 | B | 300 | 800 | 190 | 600 | 150 | 38 |
| W33x118 | B | 350 | 900 | 240 | 600 | 150 | 38 |
| W36x150 | B | 350 | 950 | 240 | 750 | 250 | 40 |
| W36x170 | B | 350 | 950 | 240 | 750 | 250 | 40 |
| W40x183 | B | 350 | 1030 | 240 | 750 | 250 | 45 |
| W6x20 | D | 200 | 200 | 120 | - | - | 20 |
| W8x35 | C | 400 | 400 | 300 | 300 | - | 20 |
| W8x35 OPTION 'A' | D | 230 | 230 | 120 | - | - | 20 |
| W10x49 | C | 450 | 450 | 350 | 350 | - | 25 |
| W10x49 OPTION 'A' | D | 280 | 280 | 150 | - | - | 25 |
| W12x72 | C | 500 | 500 | 400 | 400 | - | 35 |
| W12x106 | C | 500 | 500 | 400 | 400 | - | 35 |
| W14x159 | C | 600 | 600 | 500 | 500 | - | 40 |
| W14x193 | C | 600 | 600 | 500 | 500 | - | 40 |
| W12x72 | A | 350 | 350 | 250 | 150 | - | 30 |
| W12x106 | A | 350 | 350 | 250 | 150 | - | 30 |
| W14x159 | A | 430 | 430 | 300 | 200 | - | 35 |
| W14x193 | A | 430 | 430 | 300 | 200 | - | 35 |

W SHAPE COLUMNS



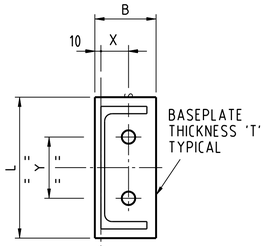

STANDARD LAYOUT



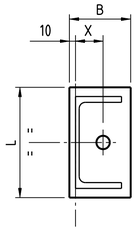

OPTION A

| MEMBER | DIMENSIONS | | | | |
|---|---|---|---|---|---|
| (LEG LENGTH) | B | L | X | Y | T |
| L3x3 | 100 | 185 | 45 | 90 | 10 |
| L3.5x3.5 | 110 | 205 | 55 | 110 | 10 |
| Lx4x4 | 130 | 225 | 55 | 110 | 12 |
| L5x5 | 150 | 265 | 75 | 150 | 16 |
| L6x6 | 180 | 310 | 90 | 180 | 16 |
| L8x8 | 230 | 410 | 120 | 240 | 20 |

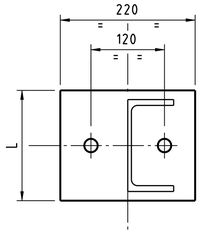
EQUAL ANGLE COLUMNS




CHANNELS ≥C8
STANDARD LAYOUT




CHANNELS ≤C6
STANDARD LAYOUT

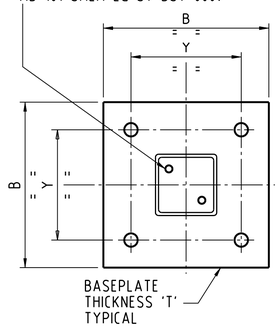



CHANNELS
OPTION A

| MEMBER | DIMENSIONS | | | | |
|---|---|---|---|---|---|
| | B | L | X | Y | T |
| C6x13 | 100 | 180 | 45 | - | 12 |
| C7x12.25 | 100 | 210 | 45 | - | 12 |
| C8x18.75 | 100 | 230 | 45 | 100 | 16 |
| C9x20 | 110 | 260 | 45 | 100 | 20 |
| C10x25 | 100 | 280 | 55 | 120 | 20 |
| C12x20.7 | 110 | 330 | 55 | 150 | 20 |
| C15x33.9 | 130 | 410 | 55 | 150 | 25 |

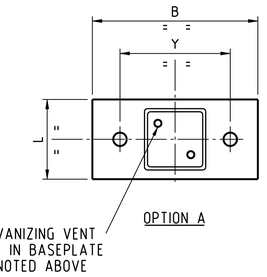
CHANNEL COLUMNS

GALVANIZING VENT HOLE IN BASEPLATE REFER NOTE 10.4 'DETAILING OF GALVANIZED STEEL' ON DRAWING MD-101-SMEM-EG-ST-DST-0009



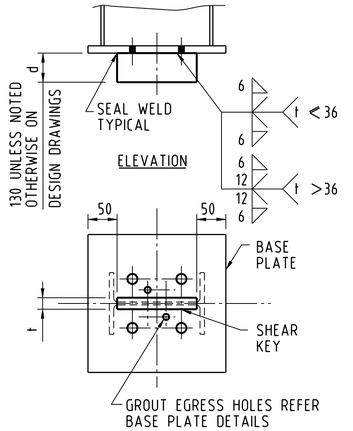

STANDARD LAYOUT

SQUARE HSS SHOWN BUT DETAIL APPLIES TO PIPE SECTIONS AS WELL



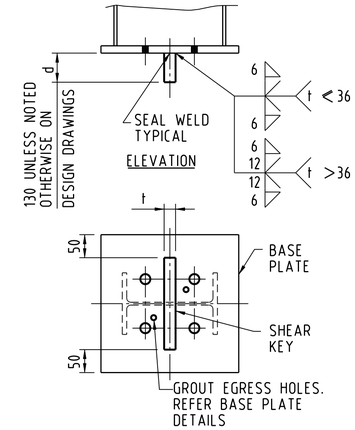

| MEMBER | | DIMENSIONS | | | | |
|---|---|---|---|---|---|---|
| SQUARE HSS | PIPE SECTION | B | L | Y | T |
| - | DN80 | 230 | 120 | 160 | 10 |
| HSS4x4 | DN100 | 270 | 130 | 180 | 10 |
| - | DN125 | 300 | 150 | 210 | 12 |
| HSS6x6 | DN150 | 340 | 180 | 240 | 12 |
| HSS8x8 | DN200 | 380 | 230 | 280 | 16 |
| HSS10x10 | DN250 | 460 | 280 | 350 | 20 |
| - | DN300 | 570 | 350 | 450 | 25 |
| - | DN350 | 600 | 380 | 480 | 25 |

SQUARE HOLLOW (HSS) & PIPE SECTIONS



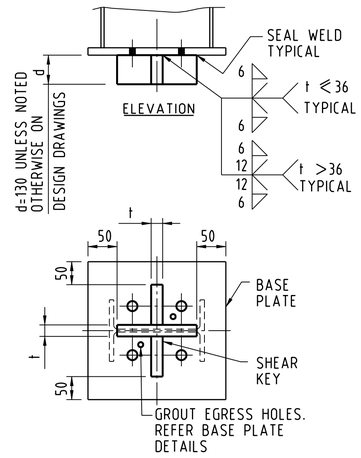

PARALLEL TO COLUMN WEB
(VIEWED FROM UNDERSIDE OF BASEPLATE)

TYPE SK1




PARALLEL TO COLUMN FLANGE
(VIEWED FROM UNDERSIDE OF BASEPLATE)

TYPE SK2




CRUCIFORM
(VIEWED FROM UNDERSIDE OF BASEPLATE)

TYPE SK3

| COLUMN SIZE | t | COLUMN SIZE | t |
|---|---|---|---|
| W8x18 | 25 | W6x20 | 25 |
| W10x22 | 25 | W8x35 | 30 |
| W10x26 | 25 | W10x49 | 32 |
| W12x26 | 30 | W12x72 | 40 |
| W14x34 | 30 | W12x106 | 40 |
| W16x40 | 30 | W14x159 | 40 |
| W18X46 | 35 | W14x193 | 40 |
| W21x50 | 35 | | |
| W24x68 | 40 | | |
| W24x84 | 40 | | |
| W27x94 | 50 | | |
| W30x116 | 50 | | |
| W33x118 | 50 | | |
| W36x150 | 50 | | |
| W36x170 | 50 | | |
| W40x183 | 50 | | |

TABLE 5

## SHEAR KEYS

**BASEPLATES** DENOTED ON DESIGN DRAWINGS

BP/A/T/D
- 'D' FOLLOWED BY A NUMBER DENOTES HOLE DIAMETER
- 'T' FOLLOWED BY A NUMBER DENOTES PLATE THICKNESS
- DENOTES OPTION WHERE APPLICABLE
- DENOTES BASEPLATE

**SHEAR KEYS** DENOTED ON DESIGN DRAWINGS

SK1/t/d
- 'd' FOLLOWED BY A NUMBER DENOTES SHEAR KEY DEPTH
- 't' FOLLOWED BY A NUMBER DENOTES PLATE THICKNESS
- DENOTES SHEAR KEY TYPE

NOTE: HOLE DIAMETER FOR BASEPLATES REFER CONCRETE DESIGN DRAWINGS & STEELWORK GENERAL NOTES DRAWING MD-101-SMEM-EG-ST-DST-0009 BOLTING NOTE 2.5

### NOTES:

1. FOR STEELWORK GENERAL NOTES REFER DRAWING MD-101-SMEM-EG-ST-DST-0009.
2. STRUCTURAL CAPACITIES OF THE CONNECTIONS SHOWN ON THIS DRAWING SHALL BE CHECKED BY THE CONTRACTOR TO DETERMINE THEIR SUITABILITY FOR THE INTENDED USE.
3. FOR TABLES REFER TO MD-101-SMEM-EG-ST-SPC-0001

## BASE PLATES

## LEGEND

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | MD |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

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REV C01 PREPARED FOR MA'ADEN BY HATCH ASSOCIATES H332701-0000-35-041-0014_C

MA'ADEN معادن Saudi Arabian Mining Company

| | | |
|---|---|---|
| DRAWN | J BADAL | 04.03.11 |
| CHECKED | J EDWARDS | 18.04.12 |
| ENGINEER | D KIRKWOOD | 18.04.12 |
| APPROVED | R COVELLA | 18.04.12 |
| MA'ADEN | | |
| SCALE NTS | | A1 |

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
BASE PLATE AND SHEAR KEY DETAILS

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0014 REV C01

$FILE$ $TIME$ $DATE$